

## Element C

### Design Requirements and Primary Stakeholder Validation:

| Priority | Design Requirement                       | Primary Stakeholder Validation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Changes color manually and automatically | Color-changing capability is important because it allows the kids to change the lighting based on their mood, or if they just want to express themselves without words. Additionally, automatically shifting through colors creates a pattern that entices and relaxes kids. The group can measure this by how many colors are available in the device, how easily the colors are changed, the rate at which they change, and how much control the kids and teachers have over them.                                                                                                             |
| 2        | Provides calming effects                 | The visual sensory device should primarily have the effect of calming down the kids when they are overstimulated. It's important that the device accomplishes this goal because sensory rooms are meant to be a space where people can calm down and engage in their five senses. Furthermore, visual sensory is one of the most important senses to stimulate because of how much it contributes to cognitive development and education. The group can measure this by asking the teachers if the kids' behavior changes after being exposed to the device, and/or how enticed they seem by it. |
| 3        | Illuminates the sensory room             | Illuminating the sensory room with various colors and designs will greatly contribute to the stimulation and "sensory" element of the classroom. Illuminations provide visual appeasement, and make the classroom look more vivid and stimulating. The group can measure this by recording the different sources of stimulating light throughout the classroom, and how much of the classroom the light encompasses (whether only a small area in the sensory room or a larger overhead light)                                                                                                   |
| 4        | Brightness can be altered                | Lights that are too bright will irritate kids more than calm them down. Also lights that are not bright enough will not keep kids engaged. Different kids will have different levels of brightness that stimulate them the most, so it's important that these levels can be changed to accommodate their differing needs. The group can measure this by how many                                                                                                                                                                                                                                 |

|   |                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                          | settings the device has for brightness, and how easily it can be changed. Additionally, the group can learn each child's brightness preference and make presets for teacher's ease of use.                                                                                                                                                                                                                                                         |
| 5 | Integrated into the room | Integrating the design into the room will help enhance the overall cohesiveness and connectivity in the sensory space. This will allow for the kids to be more immersed in the environment; therefore, boosting the overall effect of the devices in the sensory room. This can be measured based on the design compatibility within the sensory room, and by asking the teachers whether the design is too much of an intrusion in the classroom. |